

Integration Review

Table: On the first page(s) of the book there is a table of integration formulas (mostly from Calc 1). You should know (or be able to quickly derive) the first 18 formulas. Four of the formulas that I would like to highlight (some we have discussed before) are:

$$\begin{aligned}\int \sec(x) dx &= \ln|\sec(x) + \tan(x)| + C & \int \tan(x) dx &= \ln|\sec(x)| + C \\ \int \csc(x) dx &= \ln|\csc(x) - \cot(x)| + C & \int \cot(x) dx &= \ln|\sin(x)| + C\end{aligned}$$

When we are evaluating a difficult integral the antiderivative formulas from calculus 1 need to be second nature to us because our main focus will be on applying a technique(s) that we have learned in chapter 3.

General Strategy for Integration:

This process should give you an idea of a general way in which to approach the evaluation of an integral.

1. Simplify the integrand if possible. Possible ways to do this include:
 - Use algebraic manipulation
 - Use a trig identity
2. Look for an obvious substitution (i.e. a standard u -substitution)
3. If steps 1 and 2 have not led to an answer, try to decide which strategy (or strategies) that we have learned in chapter 3 looks like it might be successful. The strategies we have learned (with one new strategy included) are:
 - Integration by parts
 - If what we are integrating consists of only trig functions we may be able to use the methods of 3.2
 - Trigonometric substitution (remember the radical forms that lead to trigonometric substitution)
 - If we have a rational function consider using partial fractions
 - Complete the square
4. Keep trying

Example 1: Find $\int \frac{\sin^3(x)}{\cos(x)} dx$.

Using the ideas of 3.2 we note that $\int \frac{\sin^3(x)}{\cos(x)} dx = \int \frac{(1 - \cos^2(x)) \sin(x)}{\cos(x)} dx$.

Now, we let $u = \cos(x) \Rightarrow du = -\sin(x)dx$. Now, we finish the problem:

$$\int \frac{(1 - \cos^2(x)) \sin(x)}{\cos(x)} dx = - \int \frac{1 - u^2}{u} du = - \int \left(\frac{1}{u} - u \right) du = -\ln|u| + \frac{1}{2}u^2 + C = -\ln|\cos(x)| + \frac{1}{2}\cos^2(x) + C.$$

Example 2: Evaluate each of the following integrals

(a) $\int \frac{x^3}{\sqrt{1+x^2}} dx$

The radical in the denominator leads us to believe that a trigonometric substitution has the potential to be successful. In particular we make the following substitution: $x = \tan(\theta)$ where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

This implies that $\theta = \tan^{-1}(x)$ and $dx = \sec^2(\theta) d\theta$. Also from this substitution we obtain:

$$\frac{x^3}{\sqrt{1+x^2}} = \frac{\tan^3(\theta)}{\sqrt{1+\tan^2(\theta)}} = \frac{\tan^3(\theta)}{\sec(\theta)}.$$

Now, the integral becomes:

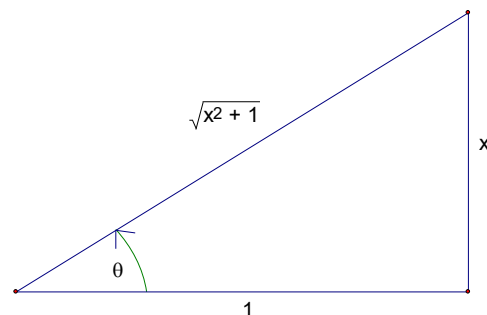
$$\int \frac{x^3}{\sqrt{1+x^2}} dx = \int \frac{\tan^3(\theta)}{\sec(\theta)} \sec^2(\theta) d\theta = \int \tan^3(\theta) \sec(\theta) d\theta = \int (\sec^2(\theta) - 1) \tan(\theta) \sec(\theta) d\theta.$$

We now make the substitution: $u = \sec(\theta) \Rightarrow du = \tan(\theta) \sec(\theta) d\theta$. Now, we have:

$$\int (\sec^2(\theta) - 1) \tan(\theta) \sec(\theta) d\theta = \int (u^2 - 1) du = \frac{1}{3} u^3 - u + C = \frac{1}{3} \sec^3(\theta) - \sec(\theta) + C.$$

The reference triangle for θ is (Note: it is the same in the first and fourth quadrants):

From this we see that: $\frac{1}{3} \sec^3(\theta) - \sec(\theta) + C = \frac{1}{3} (\sqrt{x^2+1})^3 - \sqrt{x^2+1} + C.$



(b) $\int_0^4 \frac{6z+5}{2z+1} dz$ (Hint: Begin by completing a polynomial long division)

Since the degree of the numerator equals the degree of the denominator we perform polynomial long division and we obtain: $\frac{6z+5}{2z+1} = 3 + \frac{2}{2z+1}$. Now, we can finish the problem:

$$\int_0^4 \frac{6z+5}{2z+1} dz = \int_0^4 \left(3 + \frac{2}{2z+1} \right) dz = \left(3z + \ln|2z+1| \right) \Big|_0^4 = 12 + \ln(9).$$

$$(c) \quad \int \frac{\sin^2(\theta)\cos^2(\theta) + \sin^4(\theta)\cos^2(\theta)}{1 - \sin^4(\theta)} d\theta$$

For this question it turns out that simplification leads us to something we can integrate. Specifically we have:

$$\begin{aligned} \int \frac{\sin^2(\theta)\cos^2(\theta) + \sin^4(\theta)\cos^2(\theta)}{1 - \sin^4(\theta)} d\theta &= \int \frac{\sin^2(\theta)\cos^2(\theta)[1 + \sin^2(\theta)]}{[1 - \sin^2(\theta)][1 + \sin^2(\theta)]} d\theta = \int \frac{\sin^2(\theta)\cos^2(\theta)}{\cos^2(\theta)} d\theta = \int \sin^2(\theta) \\ &= \int \frac{1}{2}[1 - \cos(2\theta)] d\theta = \frac{1}{2}\theta - \frac{1}{4}\sin(2\theta) + C \end{aligned}$$

$$(d) \quad \int_0^1 \sqrt{\frac{1-x}{1+x}} dx \quad (\text{Hint: Begin by rationalizing the numerator})$$

This is an example of an integral which is quite difficult to evaluate. It turns out that rationalizing the numerator makes it possible to evaluate:

$$\int_0^1 \sqrt{\frac{1-x}{1+x}} dx = \int_0^1 \frac{\sqrt{1-x}}{\sqrt{1+x}} \cdot \frac{\sqrt{1-x}}{\sqrt{1-x}} dx = \int_0^1 \frac{1-x}{\sqrt{1-x^2}} dx = \int_0^1 \frac{1}{\sqrt{1-x^2}} dx - \int_0^1 \frac{x}{\sqrt{1-x^2}} dx.$$

Now, we have two integrals to

evaluate. The first integrand is the derivative of inverse sine and for the second integral we make the

substitution: $u = 1 - x^2 \Rightarrow du = -2x dx \Leftrightarrow \frac{-1}{2} du = x dx$. From this substitution if we view u as function of x we

have $u(0) = 1$ and $u(1) = 0$. Thus, we have:

$$\int_0^1 \frac{1}{\sqrt{1-x^2}} dx - \int_0^1 \frac{x}{\sqrt{1-x^2}} dx = \sin^{-1}(x) \Big|_0^1 + \frac{1}{2} \int_1^0 \frac{1}{\sqrt{u}} du = \sin^{-1}(1) - \sin^{-1}(0) + \sqrt{u} \Big|_1^0 = \frac{\pi}{2} - 1.$$

Example 3: Evaluate each of the following integrals.

(a) $\int \frac{4^x + 10^x}{2^x} dx$ (Hint: Use Algebra)

Simplification is the key to this one. Notice:

$$\int \frac{4^x + 10^x}{2^x} dx = \int \frac{2^x \cdot 2^x + 2^x \cdot 5^x}{2^x} dx = \int \frac{2^x(2^x + 5^x)}{2^x} dx = \int (2^x + 5^x) dx = \frac{2^x}{\ln(2)} + \frac{5^x}{\ln(5)} + C.$$

(b) $\int_{-1}^0 \frac{x^2}{x^2 + 1} dx$ (Hint: Use Algebra)

Since the function $f(x) = \frac{x^2}{x^2 + 1}$ is continuous on $[-1, 0]$, so by the FTCP2 we know that:

$$\begin{aligned} \int_{-1}^0 \frac{x^2}{x^2 + 1} dx &= \int_{-1}^0 \frac{x^2 + 1 - 1}{x^2 + 1} dx = \int_{-1}^0 \left[\frac{x^2 + 1}{x^2 + 1} - \frac{1}{x^2 + 1} \right] dx = \int_{-1}^0 \left[1 - \frac{1}{x^2 + 1} \right] dx = \left[x - \tan^{-1}(x) \right]_{-1}^0 \\ &= [0 - \tan^{-1}(0)] - [-1 - \tan^{-1}(-1)] = [0 - 0] - \left[-1 + \frac{\pi}{4} \right] \\ &= 1 - \frac{\pi}{4} \end{aligned}$$

(c) $\int \frac{\tan^{-1}(x)}{x^2} dx$

We will use integration by parts. We let $u = \tan^{-1}(x)$ and $\frac{1}{x^2} dx = dv$. These equations imply $du = \frac{1}{1+x^2} dx$

and $\frac{-1}{x} = v$. Now, integration by parts yields:

$$\int \frac{\tan^{-1}(x)}{x^2} dx = \frac{-\tan^{-1}(x)}{x} - \int \frac{-1}{x(1+x^2)} dx = \frac{-\tan^{-1}(x)}{x} + \int \frac{1}{x(1+x^2)} dx.$$

Now we turn our focus to $\int \frac{1}{x(1+x^2)} dx$. We use partial fractions to find this. In particular we seek A , B , and C so that

$$\frac{1}{x(1+x^2)} = \frac{A}{x} + \frac{Bx+C}{1+x^2} \Rightarrow 1 = A(1+x^2) + (Bx+C)x.$$

If we let $x = 0$, we see that $A = 1$. Using this along with the equation yields: $1 = 1(1+x^2) + (Bx+C)x \Leftrightarrow 1 = 1 + x^2 + Bx^2 + Cx$. From this we conclude that $B = -1$ and

$$C = 0. \text{ This means } \int \frac{1}{x(1+x^2)} dx = \int \left(\frac{1}{x} - \frac{x}{1+x^2} \right) dx = \ln|x| - \frac{1}{2} \ln(1+x^2) + C \text{ (Note: one could also divide}$$

this into two integrals and use a simple substitution in one of the integrals).

$$\text{Finally, we have: } \frac{-\tan^{-1}(x)}{x} + \int \frac{1}{x(1+x^2)} dx = \frac{-\tan^{-1}(x)}{x} + \ln|x| - \frac{1}{2} \ln(1+x^2) + C.$$